Thus the way was paved for a quantitative determination of the weight or mass ratios of atoms. Although the absolute number of atoms present at any one time was not known, it was known for a certainty that at the same temperature and pressure, the number of molecules contained in equal volumes of gases was the same. This was sufficient, since it furnished information concerning the mass ratios of atoms and molecules.

Not much later, the Swedish scientist Berzelius determined the atomic weights of a great many molecules, and succeeded in developing very definite theories about the way in which molecules are built from individual atoms. Berzelius also studied the nature of the forces binding atoms together in molecules. It was he who introduced the notion of *valency* in connection with the force binding an atom of one element to an atom of another. In studying this force, he came to the conclusion that it must be of electrical nature.

The status of atomic theory about 120 years ago can be summed up as follows: It was known that the prodigious number of chemical compounds could be reduced to a relatively small number of chemical elements, a great many—although by far not all—of which were known. The mass ratios of the atoms of these elements were also known fairly accurately—as, for instance, that one oxygen atom was roughly 16 times, and a nitrogen atom roughly 14 times heavier than a hydrogen atom. However, there were still considerable gaps to be filled. Nothing at all was known about the absolute size of atoms, or the order of magnitude of their number within. a given volume of space. All that was known was that in gases at the same temperature and pressure, the number of molecules was the same. So far as accurate knowledge was concerned, an atom might still, as Democritus had believed, be approximately the same size as one of the motes dancing in a sunbeam, or infinitely smaller. Just as little was known about the shape of atoms, or about the forces operating between them. As to the latter question, nothing but extremely hypothetical conjectures was possible. Furthermore, although it was known that, chemically, the atoms must be the ultimate building blocks of matter—in other words, the smallest units demonstrable by chemical means and methods—no one knew whether or not these chemical atoms might be capable of being further subdivided and transmuted into each other by the application of other methods.

A discovery, from which Prout was the first to draw conclusions in 1815, actually spoke against the theory of the absolute indivisibility of atoms. Prout (1785–1850) based his deductions on the fact that the atomic weights known in those days—these were mainly only those of the lighter elements—were nearly integral multiples of the atomic weight of hydrogen. This fact is the basis of his view, that all atoms were built up of hydrogen atoms. Since one carbon atom was roughly twelve times, and one oxygen atom roughly sixteen times as heavy as one hydrogen atom, the carbon atom had to be composed of